

LUCAS FRERY

Electronics Hardware Engineer

lucas.frery@gmail.com | +33 7 81 70 23 52 | [Github](#) | [linkedin](#) | [e-Portfolio](#) | Open to international relocation

Electronics Engineer experienced designing **PCBs, PCBAs and embedded systems** from electrical system architecture and component selection through schematic design, PCB layout, rapid prototyping, hardware bring-up, characterization and validation. Hands-on experience with microcontrollers, mixed-signal I/O, power distribution and protection circuitry, communication interfaces, EMI/EMC, signal integrity and hardware debugging. Interested in robotics, drones, autonomous systems and defence. **Available from October 2026.**

CORE STRENGTH

Schematic Capture & PCB Design · Embedded Systems · STM32 / Microcontrollers · C / C++ / Python · Mixed-Signal Electronics · Power Distribution & Protection Circuitry · Rapid Prototyping & Iterative Testing · Hardware Bring-up, Debugging & Validation · Signal Integrity / EMI/EMC · CAN / LIN / Ethernet / SPI / I2C / UART · Electrical System Integration

WORK EXPERIENCE

Electronics Hardware Engineer

Mar 2026 – Sep 2026

renco GmbH

Cologne, Germany

- Designed a compact **hardware-in-the-loop** platform for automotive VCU testing and validation, integrating analog/digital I/O and a **dual-STM32** architecture communicating over **SPI at 42 Mbit/s**, with **CAN, LIN, Ethernet, SPI, I2C, USB and USART** interfaces.
- Delivered the complete system schematics in **Altium Designer** by performing component selection and engineering trade-offs across electrical performance, cost, availability and integration constraints, supported by hierarchical design, ActiveBOM and ERC.
- Designed 4-layer PCB **layouts** considering controlled impedance, length matching, signal integrity, EMI/EMC, return-current paths, DRC and DFM practices.
- Reduced full-system integration risk by designing, assembling and testing dedicated prototype PCBAs for hardware **bring-up**, unit-level validation, electrical characterization and root-cause troubleshooting before release of the complete platform.

Hardware/Embedded Systems Engineer

Sep 2023 – Jan 2026

ESTACARS – Formula Student

Laval, France

- Designed, simulated, and validated safety-critical **PCBAs** for an electric race car, covering schematic design, signal conditioning, and system integration into the car (**KiCad, LTSpice**).
- Resolved **EMI issues** in RS-485 BMS communication through hardware filtering, grounding strategy, and signal-integrity analysis, improving system robustness and reliability.
- Commissioned and calibrated the **electric powertrain** (inverter, motor, sensors), defining safe operating limits, troubleshooting, and validation procedures.
- Performed system-level **bring-up, testing, and root-cause analysis** during vehicle integration and track testing, ensuring reliable operation under real-world conditions.

R&D Engineer

May 2025 – Oct 2025

Involute Transmissions

Caen, France

- Developed **computational models** to predict rolling-element bearing lifespan, power losses, deflections, stresses and stiffness under various operating conditions.
- Designed and programmed a **Scilab-based engineering tool with GUI** to automate bearing performance calculations and support engineering decision-making.
- Validated simulation results against commercial software, scientific literature and **ISO standards** to ensure model accuracy.

EDUCATION

Master of Engineering

Expected Fall 2026

ESTACA – Graduate School of Engineering

Laval, France

- Specialized in Automotive Embedded Systems and Electrical Engineering

SKILLS

Electronics: Electrical System Architecture, Schematic Capture, PCB Design & Layout, Power Distribution, Protection Circuitry, Wiring Harness Design, EMI/EMC, Signal Integrity, Hardware Bring-up, Debugging & Validation

Embedded Systems & Interfaces: STM32, Microcontrollers, RS-485, SPI, I2C, UART/USART, CAN, LIN, Ethernet, USB, PWM, ADC/DAC

EDA Tools: Altium Designer, KiCad, LTSpice

Programming: C, C++, Python, MATLAB/Simulink, LaTeX

Mechanical & FEA: SolidWorks, CATIA, 3DEXPERIENCE

Languages: French native, English fluent, German elementary, Spanish elementary